



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2023

UE21CS242A - Web Technologies

Total Marks : 100.0

1.a. What are the 3 ways to include CSS styling in CSS. explain with examples (6.0 Marks)

1.b. Write the code to get the following output using tables

Student details

Sl num	Name	Project Title
1	Rahul	Gaming Web
2	Rakul	E-Commerce
3	Mahi	Web Hack
4	virat	

(6.0 Marks)

1.c. Write a code to get the following code

Fill your details

Personal Info

Name

LastName

Enter your email id

Password

Enter your address

Select Gender

Gender

Select Gender

☐ Male

☐ Female

Select fav lang

Fav lan

☐ HTML

☐ C++

☐ Java

Submit

Reset

(8.0 Marks)

2.a. What are the different ways to get an element from DOM? Explain any 3 with an example (5.0 Marks)

2.b. Write a JavaScript program to calculate multiplication and division of two numbers to get output as below.

(6.0 Marks)

1st Number :

2nd Number:

The Result Is :
120

2.c. Write jquery code to get the following output

hello students
hello pes
web tech

when user clicks Hide button text should hide when user clicks show button text should get displayed. (5.0 Marks)

2.d. Write difference between XML and JSON (4.0 Marks)

3.a. Explain the following methods of component life cycle:

- render()
- componentDidMount
- componentDidUpdate
- componentWillUnmount

(4.0 Marks)

3.b. write difference between Stateless and Stateful components in React
(4.0 Marks)

3.c. Write a code to get the following output in browser using React map and keys

(6.0 Marks)

- Apple
- Mango
- Pineapple
- Cherry

3.d. Write a code to get the following output in react using components, Create a class component called Article1 which is having states such as name,color and price if user clicks change color button text red should get changed blue.

my car color is red

price is 10000000

Change color

(6.0 Marks)

4.a. Write in Breif important features of Node.JS

(4.0 Marks)

4.b. What are buffers in Node JS? Write a program to create a buffer,write data to buffer and read data from buffer (4.0 Marks)

4.c. Write user defined module in Node js server called calc.js which contains different 3 functions such as

- 1..to return addition of 2 numbers
- 2.to return subtraction of 2 numbers
- 3.to return Multiplication of 2 numbers

write a code to include calc.js in node.js file and print all result in console
(6.0 Marks)

4.d. Write code snippet to connect to a db named "Groceries" on a Mongodb instance running on localhost:27017 and write the following details into the "Fruits" collection,insert many documents(atleast 5) with details: Fruit-name and price.
It should handle errors appropriately.
Note: write only the code required to write into the database. (6.0 Marks)

5.a. Explain the following

- 1.What is Web Service?
- 2.Why Web Services are useful?
- 3.What is REST API?

(4.0 Marks)

5.b. what are middleware functions? Write briefly about the order of execution of middleware functions (4.0 Marks)

5.c. Write a code in express to create a server which have 3 different route paths like /pes,/pes/:branch,/pes/branch/id which returns 3 different contents when particular route has been entered in browser if user enters following paths server should return following contents
/pes → Hello PES should be returned from the server
/pes/:CSE → branch name is CSE
/PES/:CSE/:123 → id:123 and name :CSE (6.0 Marks)

5.d. Write a route called "picsupload" which receives multiple image files in a POST request. The route function should save each of them under the "pics" folder in the same directory as the JS file. Any error like, no files uploaded or saving files failed should be handled accordingly.
Note: Write only the server JS code. HTML/client code is not required. (6.0 Marks)